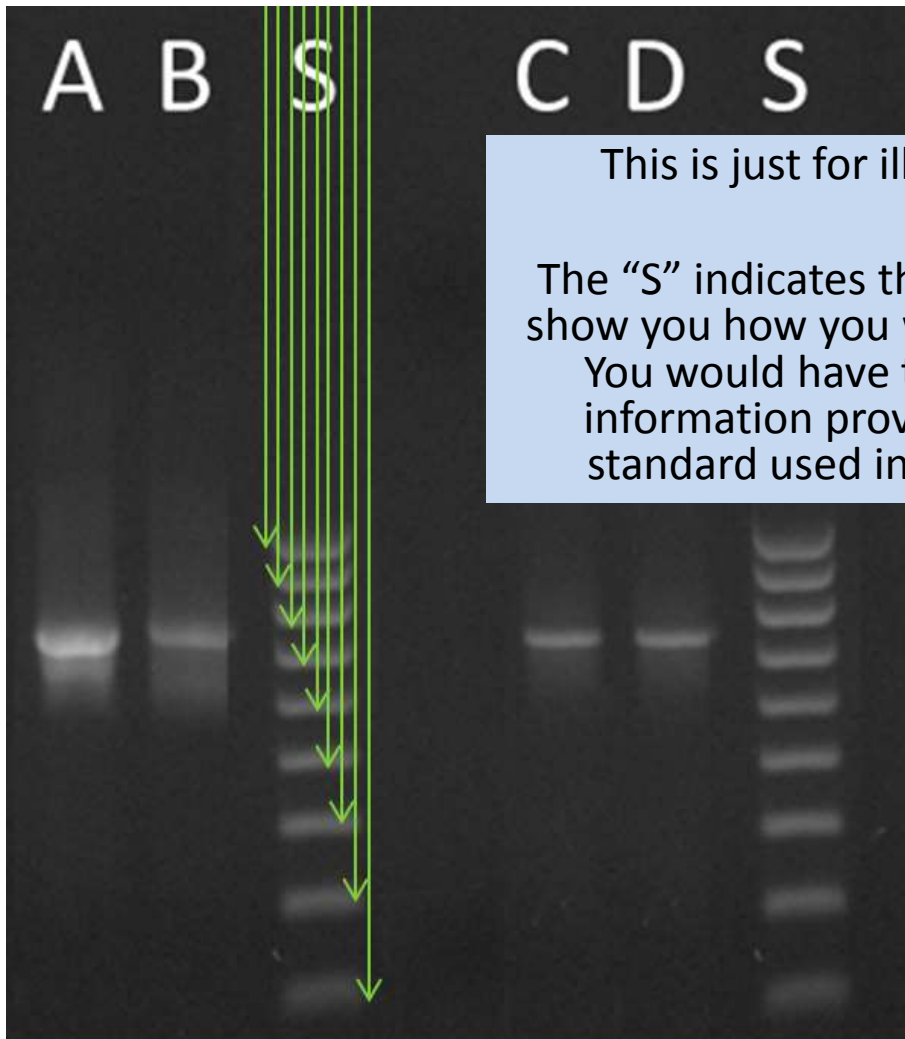


# How to Make and Use a Standard Curve To Determine the Size (in bp) of a DNA fragment on a Gel

# Steps

1. Print the picture of the gel on paper and get a ruler and a pencil.
2. Look at the lane that contains the standard for the gel. Identify the size of each standard band in bp. Enter that data into a column in Excel.
3. Right beside that (in the next column) determine the log (base 10) of the band sizes. =LOG10(cellref). **THIS IS YOUR Y DATA**
4. Measure the distance (in cm) from the top of the gel to each band in the ladder. Enter that data into a column to the left of your Y data. This will be your X data.
5. Graph your X and Y data. Use a scatterplot. Fit a trendline (linear) and show the  $y=mx+b$  equation and the  $R^2$  value on your graph. Note that your graph doesn't have to have a figure title and caption, but does need the axes correctly labelled.
6. Measure the distance migrated (again, in cm) for the unknown sample (you will have only one on a practice gel image). Use the standard curve to calculate Y, given that those distances are the X values.
7. Take the antilog of each to solve for the size of the unknown bands in bp. Excel can do that using the equation =10^(cellref).



This is just for illustration! Do not print out or USE this gel.

The “S” indicates the molecular weight standard. The green lines show you how you would measure them to determine the X data. You would have to know the size of each band based on the information provided by the supplier (this is a picture of the standard used in your lab, it ranges from 900 bp to 100 bp).

# What the Excel Workbook Should Basically Look Like

|    | A | B                     | C                             | D            | E |
|----|---|-----------------------|-------------------------------|--------------|---|
| 1  |   | Standard Curve        |                               |              |   |
| 2  |   | X (distance migrated) | Size of Ladder Fragment in BP | Y LOG10 Size |   |
| 3  |   |                       | 900                           | =LOG10(C3)   |   |
| 4  |   |                       | 800                           |              |   |
| 5  |   | What you              | 700                           |              |   |
| 6  |   | measured              | 600                           |              |   |
| 7  |   | with your             | 500                           |              |   |
| 8  |   | ruler goes            | 400                           |              |   |
| 9  |   | here.                 | 300                           |              |   |
| 10 |   |                       | 200                           |              |   |
| 11 |   |                       | 100                           |              |   |
| 12 |   |                       |                               |              |   |
| 13 |   |                       |                               |              |   |
| 14 |   |                       |                               |              |   |
| 15 |   |                       |                               |              |   |

Again – do not use this data – it is to teach you about making a standard curve.

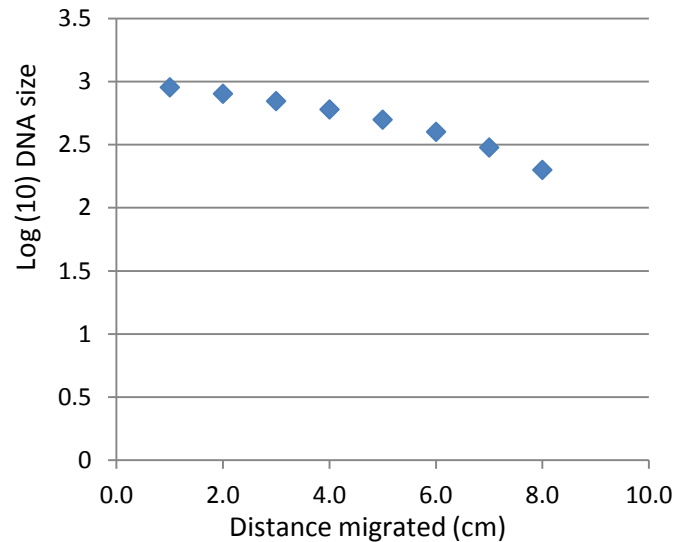
Above is the equation you would type into Excel (note the cell reference) to calculate the  $\log(10)$  of the DNA size.

demotablestand

| File Home Insert Page Layout Formulas Data Review View |                           |                               |              |           |
|--------------------------------------------------------|---------------------------|-------------------------------|--------------|-----------|
| Clipboard                                              |                           | Font                          |              | Alignment |
| D4 $\text{f}_x$ =LOG10(C4)                             |                           |                               |              |           |
| A                                                      | B                         | C                             | D            | E         |
| 1                                                      | Making the Standard Curve |                               |              |           |
| 2                                                      | This will be the X data   |                               |              |           |
| 3                                                      | Distance migrated (cm)    | Size of Ladder Fragment in BP | Y LOG10 Size |           |
| 4                                                      | 6.6                       | 900                           | 2.954242509  |           |
| 5                                                      | 7                         | 800                           | 2.903089987  |           |
| 6                                                      | 7.5                       | 700                           | 2.84509804   |           |
| 7                                                      | 7.8                       | 600                           | 2.77815125   |           |
| 8                                                      | 8.3                       | 500                           | 2.698970004  |           |
| 9                                                      | 8.9                       | 400                           | 2.602059991  |           |
| 10                                                     | 9.5                       | 300                           | 2.477121255  |           |
| 11                                                     | 10.3                      | 200                           | 2.301029996  |           |
| 12                                                     | 11.5                      | 100                           | 2            |           |
| 13                                                     |                           |                               |              |           |
| 14                                                     | The data above was        |                               |              |           |
| 15                                                     | measured with ruler       |                               |              |           |
| 16                                                     | and pencil on the         |                               |              |           |
| 17                                                     | print out of the gel.     |                               |              |           |
| 18                                                     |                           |                               |              |           |
| 19                                                     |                           |                               |              |           |

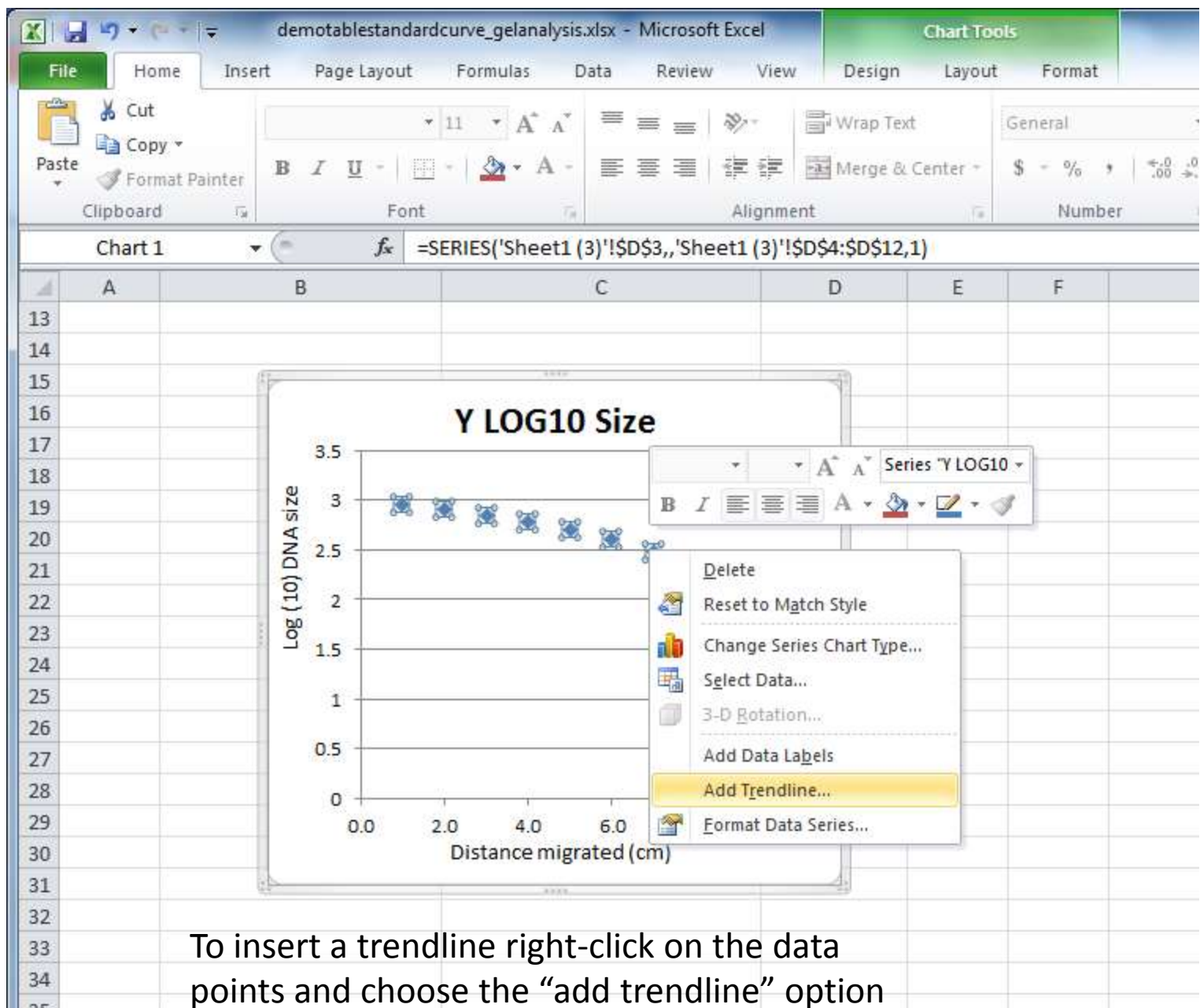
This is example data  
for you to look at  
(not use).

This is an example of  
a complete Y data  
Column (column D)

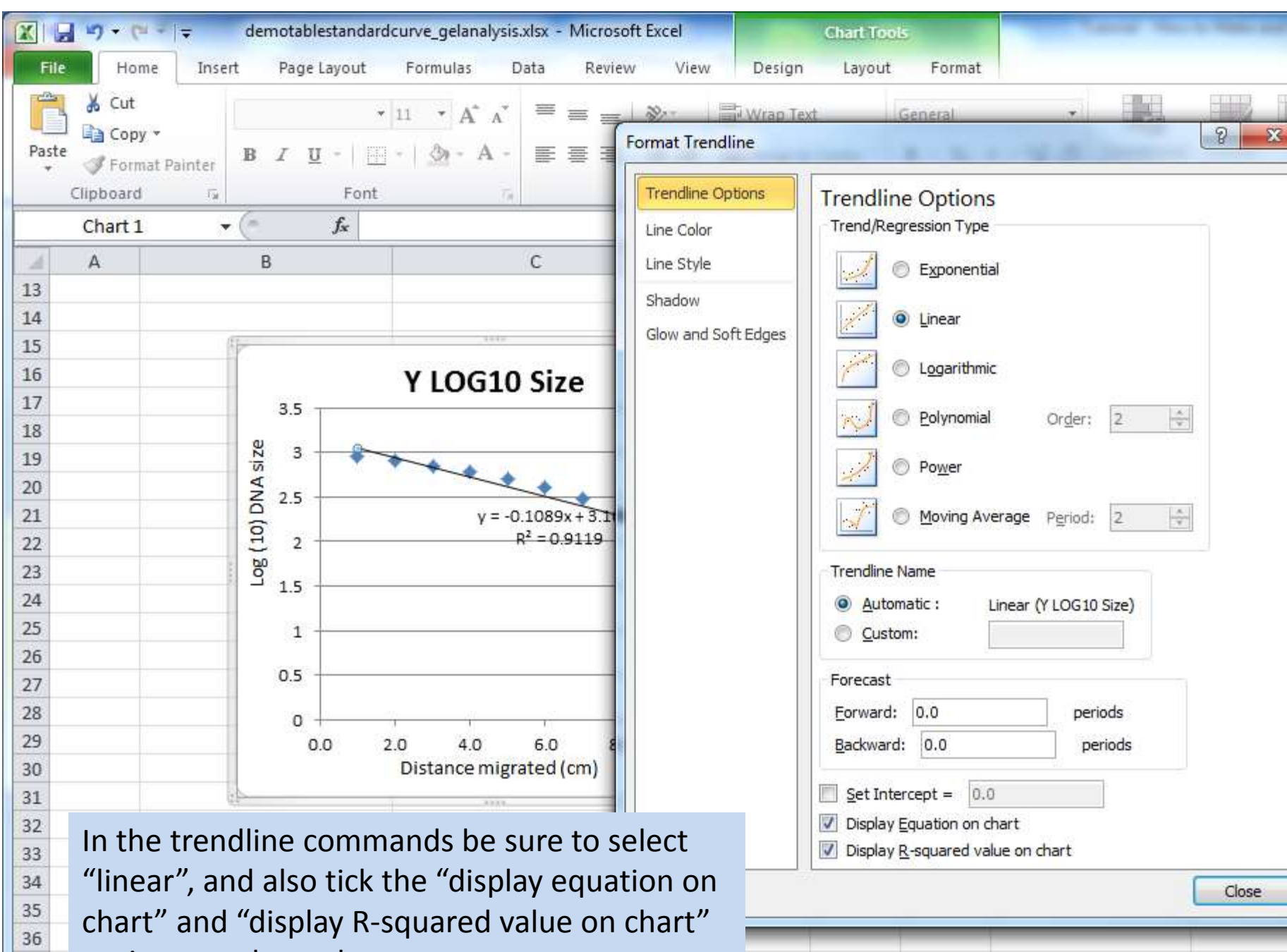


Next, you make a scatterplot style graph, like the one shown (it will slope down and right). Be sure to label your axes as shown.

If Excel doesn't place the X and Y data as shown, right click in the graph itself and use the "select data source" area to change how it is plotting.

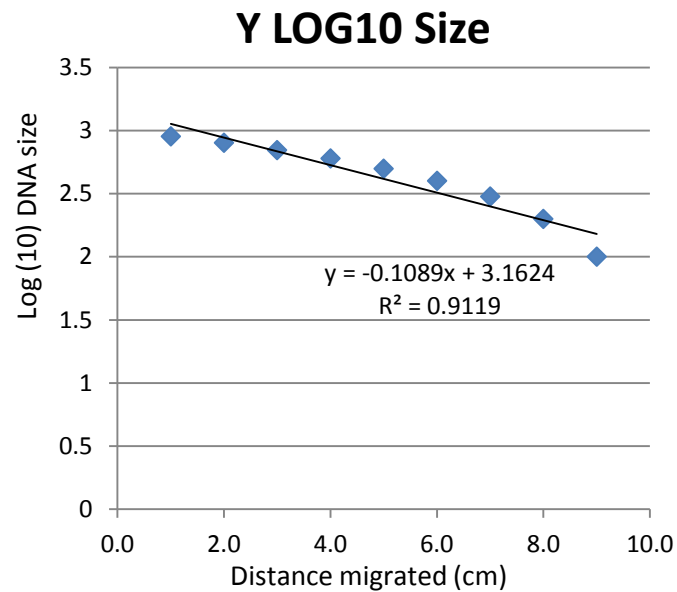


To insert a trendline right-click on the data points and choose the “add trendline” option as shown.



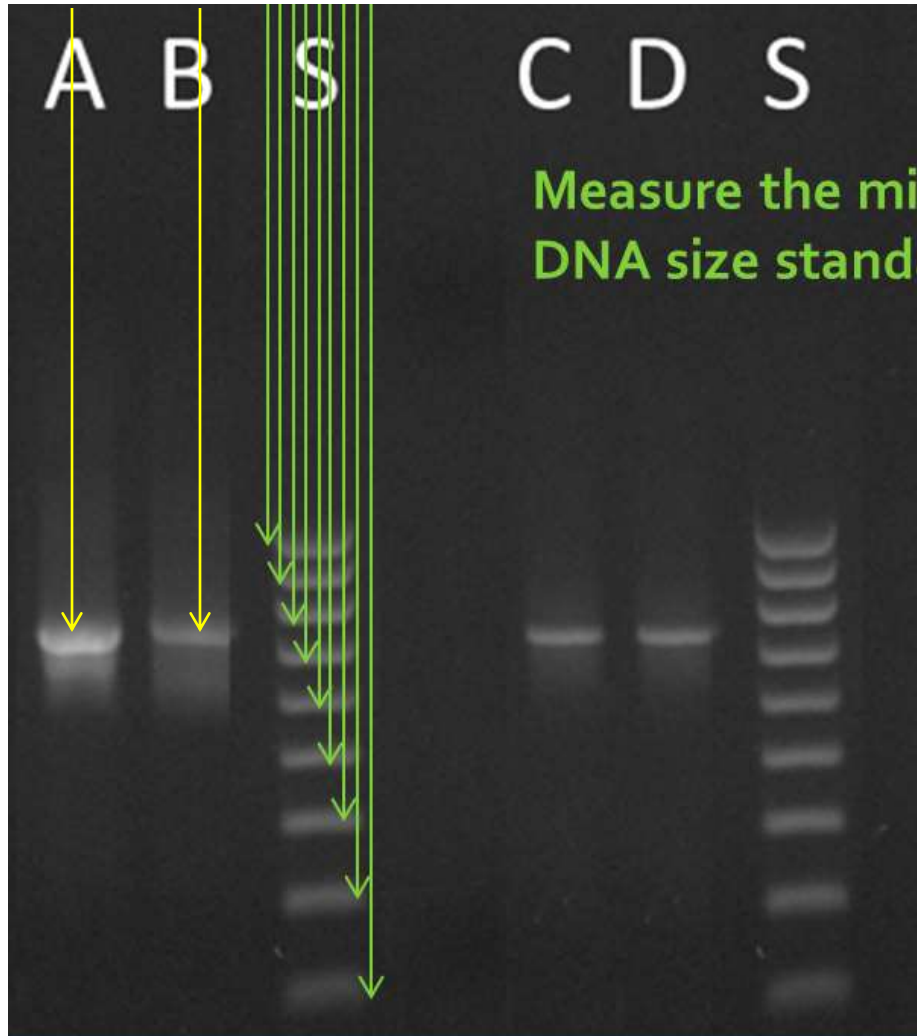
In the trendline commands be sure to select “linear”, and also tick the “display equation on chart” and “display R-squared value on chart” options, as shown here.





You should now have the standard curve equation, and a measure of the quality of the fit. This one is not ideal (ideal is 1.00) but we can live with it for the purposes of illustration!

# Using the Standard Curve – an EXAMPLE ONLY



1. Measure the distance migrated (in cm) for each of your two unknown samples (your PCR products). In the picture on the left that would be the yellow lines. Plug these into the standard curve (they are the X values), one at a time. Solve for Y.
2. Take the antilog of each calculated Y to solve for the size of the unknown bands in bp. Excel can do that using the equation  $=10^{(\text{cellref})}$ .
3. Do a common sense check. In this example if you calculated something wildly different from 650 bp it would be odd (because look at how the unknowns run...).

Solving for Y.

If my standard curve is:  $y = -0.1089x + 3.1624$

And the distance I measured to the first band is 7.6 cm....then I use 7.6 as the X data and solve for y.

$$\begin{aligned} Y &= -0.1089(7.6) + 3.1624 \\ &= -2.33476 \end{aligned}$$

That's obviously not a DNA size in BP – because I have to take the ANTILOG of that number!

In Excel that command is  $=10^{(2.33476)} = 216 \text{ bp}$

## **Critiquing what you got.**

When I just look at the gel, I can see that my standard curve isn't great! (Because the bands of interest actually run between 600 and 700 bp distances). So why is my calculated size only 216 bp?

- Answer - The bands are fuzzy in the samples and the standards, making the measurements less than perfect.
- Solution – redo the analysis and optimize gel conditions to get crisper looking bands.